CV QSDC Gaussian-States Protocol Method

1. Bob prepares squeezed coherent states

For each time slot $j \in \{1, \ldots, n\}$, Bob samples

$$\alpha_j = q_j + ip_j, \qquad q_j, p_j \in \mathbb{R},$$

and

$$s_j = r_j e^{i\theta_j}, \qquad r_j \geq 0, \quad \theta_j \in [0, 2\pi).$$

Bob prepares

$$|\psi_j\rangle = S(s_j)D(\alpha_j)|0\rangle.$$

The displacement and squeezing operators are

$$D(\alpha) = \exp(\alpha\hat{a}^\dagger - \alpha^*\hat{a}),$$

$$S(s) = \exp\left[\frac{1}{2}\left(s^*\hat{a}^2 - s\hat{a}^{\dagger 2}\right)\right].$$

The operator identity used throughout the protocol is

$$S(s)D(\alpha) = D(\beta)S(s),$$

where

$$\beta = \alpha\cosh r + e^{i\theta}\alpha^*\sinh r.$$

The inverse relation is

$$\alpha = \beta\cosh r - e^{i\theta}\beta^*\sinh r.$$

2. Alice checks the control modes

Alice selects $n/2$ received states as control modes. Bob reveals α_j and s_j for those coordinates.

For each control state, Alice applies

$$S^\dagger(s_j) = S(-s_j),$$

so that

$$S(-s_j)S(s_j)D(\alpha_j)|0\rangle = D(\alpha_j)|0\rangle.$$

Alice homodyne-measures one quadrature:

$$q_j = \mathrm{Re}(\alpha_j) \quad \text{or} \quad p_j = \mathrm{Im}(\alpha_j).$$

The protocol continues only if the measured value matches the corresponding component of α_j within the chosen tolerance.

3. Alice encodes message modes

The real line is partitioned as

$$(-\infty,-3),\ (-3,-2),\ (-2,-1),\ (-1,0),\ (0,1),\ (1,2),\ (2,3),\ (3,\infty),$$

corresponding to

$$000,\ 001,\ 010,\ 011,\ 100,\ 101,\ 110,\ 111.$$

For a 3-bit symbol, Alice samples m_A from the corresponding interval and sets

$$\beta_A = m_A(1+i).$$

On a message state, Alice applies

$$D(\beta_A).$$

The remaining non-control states are left unchanged as decoys:

$$D(0) = I.$$

4. Bob checks decoys and decodes messages

Bob applies

$$S(-s_j)$$

to each returned non-control state.

For a decoy state,

$$S(-s_j)S(s_j)D(\alpha_j)|0\rangle = D(\alpha_j)|0\rangle,$$

so Bob checks the measured quadrature value against the corresponding component of α_j.

For a message state, Alice's displacement appears after Bob unsqueezes as

$$\Delta_j = \beta_A \cosh r_j - e^{i\theta_j}\beta_A^* \sinh r_j.$$

Bob subtracts his original displacement,

$$\Delta_j = \alpha_{\mathrm{meas},j} - \alpha_j,$$

and recovers Alice's displacement with

$$\beta_A = \Delta_j \cosh r_j + e^{i\theta_j}\Delta_j^* \sinh r_j.$$

Since

$$\beta_A = m_A(1+i),$$

Bob estimates

$$m_A = \frac{\mathrm{Re}(\beta_A) + \mathrm{Im}(\beta_A)}{2},$$

then maps m_A back to its interval and obtains the corresponding 3-bit symbol.